

OptiCrop: Project Planning

Milestones

Milestone	Deliverable
Problem definition & data collection	Problem statement, dataset selected (Crop_recommendation.csv)
Exploratory Data Analysis	EDA notebook: distributions, correlations, outlier handling
Preprocessing & feature engineering	Scaled features, encoded labels, seasonal grouping logic
Model building & evaluation	Logistic Regression model, accuracy/precision/recall/F1/ROC-AUC
Clustering analysis	K-Means clustering (elbow method, k=4) for soil/climate profiling
Application development	Flask app with input form and result page
Testing	Manual functional and performance testing of the web app
Documentation & demo	README, ERD, architecture diagram, demo recording

Task Breakdown Approach

Work was organized around the standard ML project lifecycle: understand the problem, prepare data, build and evaluate a model, then wrap it in a usable interface, followed by testing and documentation.